1 The iterated conditioning property

Suppose $\mathcal{F}$ and $\mathcal{G}$ are two σ-algebras such that $\mathcal{G} \subseteq \mathcal{F}$, that is, $\mathcal{G}$ is a sub σ-algebra of $\mathcal{F}$. In this section, we prove the iterated conditioning property that involve σ-algebras which are countably generated, i.e., are generated by a finite or countable collection of events of the sample space. The iterated conditioning property reads

$$\mathbb{E}\big[\mathbb{E}[\mathbb{X} \mid \mathcal{F}] \mid \mathcal{G}\big] = \mathbb{E}[\mathbb{X} \mid \mathcal{G}] \ . \tag{1}$$

Exercise 1a (2 points): Consider the σ-algebras $\mathcal{F}_1$ and $\mathcal{F}_2$ of the first two and first three coin tosses respectively. (Take $N = 3$ for this exercise.) Show that $\mathcal{F}_1$ is a sub σ-algebra of $\mathcal{F}_2$ by arguing that every event in $\mathcal{F}_1$ is present in $\mathcal{F}_2$.

Ans. Here $\mathcal{F}_1$ and $\mathcal{F}_2$ are the σ-algebras determined by the first one and the first two coin tosses respectively. The sample space is

$$\Omega_3 = \{HHH, HHT, HTH, HTT, THH, THT, TTH, TTT\} \ . \tag{2}$$

The events of $\mathcal{F}_1$ are exactly those whose occurrence is decided by ω_1 alone:

$$\begin{aligned} &\mathcal{F}_1 = \big\{\emptyset,\ \Omega_3,\ A_H,\ A_T\big\} \ , \\ &A_H = \{HHH, HHT, HTH, HTT\} \ , \quad A_T = \{THH, THT, TTH, TTT\} \ . \end{aligned} \tag{3}$$

The events of $\mathcal{F}_2$ are those decided by $\omega_1\omega_2$, i.e., all events generated by unions of some or all of the four sets

$$\begin{aligned} A_{HH} &= \{HHH, HHT\} \ , \quad A_{HT} = \{HTH, HTT\} \ , \\ A_{TH} &= \{THH, THT\} \ , \quad A_{TT} = \{TTH, TTT\} \ , \end{aligned} \tag{4}$$

giving $2^4 = 16$ events. Now

$$\begin{aligned} &\emptyset \in \mathcal{F}_2 \ , \qquad \Omega_3 = A_{HH} \cup A_{HT} \cup A_{TH} \cup A_{TT} \in \mathcal{F}_2 \ , \\ &A_H = A_{HH} \cup A_{HT} \in \mathcal{F}_2 \ , \qquad A_T = A_{TH} \cup A_{TT} \in \mathcal{F}_2 \ . \end{aligned} \tag{5}$$

Every event of $\mathcal{F}_1$ is therefore an event of $\mathcal{F}_2$, so $\mathcal{F}_1 \subseteq \mathcal{F}_2$ and $\mathcal{F}_1$ is a sub σ-algebra of $\mathcal{F}_2$. ■

Exercise 1b (2 points): List the atoms of $\mathcal{F}_1$ and $\mathcal{F}_2$ and show that the atoms of $\mathcal{F}_1$ are events in $\mathcal{F}_2$, and give the composition of the atoms of $\mathcal{F}_1$ in terms of the atoms of $\mathcal{F}_2$.

Ans. The atoms (the smallest non-empty events) for $\mathcal{F}_1$ and $\mathcal{F}_2$ are

$$\begin{aligned} \mathcal{F}_1 :&\quad A_H = \{HHH, HHT, HTH, HTT\} \ , \quad A_T = \{THH, THT, TTH, TTT\} \ , \\ \mathcal{F}_2 :&\quad A_{HH} = \{HHH, HHT\} \ , \ A_{HT} = \{HTH, HTT\} \ , \\ &\quad A_{TH} = \{THH, THT\} \ , \ A_{TT} = \{TTH, TTT\} \ . \end{aligned}$$

The atoms of $\mathcal{F}_1$ are events in $\mathcal{F}_2$ and they are the following unions of atoms of $\mathcal{F}_2$:

$$A_H = A_{HH} \cup A_{HT} \ , \qquad A_T = A_{TH} \cup A_{TT} \ . \tag{6}$$

In general, $A_{\omega_1} = A_{\omega_1 H} \cup A_{\omega_1 T}$. ■

Recall that the conditional expectation random variable $\mathbb{E}[\mathbb{X} \mid \mathcal{F}]$ is defined in the following way when the atoms of $\mathcal{F}$ generate it. The value of $\mathbb{E}[\mathbb{X} \mid \mathcal{F}]$ on a particular element $\underline{\omega}$ of sample space is given by the conditional expectation value $\mathbb{E}[\mathbb{X} \mid A^{\underline{\omega}}]$ where $A^{\underline{\omega}}$ is the unique atom in $\mathcal{F}$ that contains $\underline{\omega}$. This conditional expectation value simply averages the value of $\mathbb{X}$ over the outcomes in the atom $A^{\underline{\omega}}$.

For instance, if we take $\mathcal{F} = \mathcal{F}_2$, the value of $\mathbb{E}[\mathbb{X} \mid \mathcal{F}_2]$ on HHH is given by $\mathbb{E}[\mathbb{X} \mid A_{HH}]$ since $A_{HH} = \{HHH, HHT\}$ is the atom in $\mathcal{F}_2$ which contains HHH. Then,

$$\begin{aligned} \mathbb{E}[\mathbb{X} \mid \mathcal{F}_2](HHH) = \mathbb{E}[\mathbb{X} \mid A_{HH}] &= \sum_{\underline{\omega} \in A_{HH}} \mathbb{X}(\underline{\omega}) \frac{\mathbb{P}\{\underline{\omega}\}}{\mathbb{P}\{A_{HH}\}} \\ &= \mathbb{X}(HHH)\frac{p^3}{p^2} + \mathbb{X}(HHT)\frac{p^2 q}{p^2} = p\mathbb{X}(HHH) + q\mathbb{X}(HHT) \ . \end{aligned} \tag{7}$$

Exercise 1c (2 points): Show that the explicit expression for the conditional expectation $\mathbb{E}[\mathbb{X} \mid \mathcal{F}_2]$ on a generic element $\omega_1\omega_2\omega_3$ of sample space is given by

$$\mathbb{E}[\mathbb{X} \mid \mathcal{F}_2](\omega_1\omega_2\omega_3) = p\mathbb{X}(\omega_1\omega_2 H) + q\mathbb{X}(\omega_1\omega_2 T) \ . \tag{8}$$

Ans. The atom of $\mathcal{F}_2$ containing $\omega_1\omega_2\omega_3$ is $A_{\omega_1\omega_2} = \{\omega_1\omega_2 H, \omega_1\omega_2 T\}$. The conditional probabilities $\mathbb{P}\{\underline{\omega} \mid A_{\omega_1\omega_2}\}$ are zero when $\underline{\omega} \notin A_{\omega_1\omega_2}$. For $\underline{\omega} \in A_{\omega_1\omega_2}$, we have

$$\mathbb{P}\{\omega_1\omega_2 H \mid A_{\omega_1\omega_2}\} = \frac{\mathbb{P}\{\omega_1\omega_2 H\}}{\mathbb{P}\{A_{\omega_1\omega_2}\}} \ , \quad \mathbb{P}\{\omega_1\omega_2 T \mid A_{\omega_1\omega_2}\} = \frac{\mathbb{P}\{\omega_1\omega_2 T\}}{\mathbb{P}\{A_{\omega_1\omega_2}\}} \ . \tag{9}$$

Since the coin tosses are independent, we have $\mathbb{P}\{A_{\omega_1\omega_2}\} = p^{n_H(\omega_1\omega_2)} q^{n_T(\omega_1\omega_2)}$. This finally gives

$$\mathbb{P}\{\omega_1\omega_2 H \mid A_{\omega_1\omega_2}\} = p \ , \quad \mathbb{P}\{\omega_1\omega_2 H \mid A_{\omega_1\omega_2}\} = q \ . \tag{10}$$

Averaging $\mathbb{X}$ over the atom,

$$\begin{aligned} \mathbb{E}[\mathbb{X} \mid \mathcal{F}_2](\omega_1\omega_2\omega_3) &= \mathbb{E}[\mathbb{X} \mid A_{\omega_1\omega_2}] \\ &= \mathbb{X}(\omega_1\omega_2 H)\mathbb{P}\{\omega_1\omega_2 H \mid A_{\omega_1\omega_2}\} + \mathbb{X}(\omega_1\omega_2 T)\mathbb{P}\{\omega_1\omega_2 T \mid A_{\omega_1\omega_2}\} \\ &= p\mathbb{X}(\omega_1\omega_2 H) + q\mathbb{X}(\omega_1\omega_2 T) \ . \end{aligned} \tag{11}$$

Note that the right hand side does not depend on the outcome of ω_3 since the two outcomes are averaged over. For $\omega_1\omega_2 = HH$ this reproduces the computation shown above. ■

Exercise 1d (2 points): Take the conditional expectation of the above explicit form of $\mathbb{E}[\mathbb{X} \mid \mathcal{F}_2]$ with respect to $\mathcal{F}_1$. Show that $\mathbb{E}\big[\mathbb{E}[\mathbb{X} \mid \mathcal{F}_2] \mid \mathcal{F}_1\big]$ is given explicitly by

$$\begin{aligned}\mathbb{E}\big[\mathbb{E}[\mathbb{X} \mid \mathcal{F}_2] \mid \mathcal{F}_1\big](\omega_1\omega_2\omega_3) \\ = p\big(p\mathbb{X}(\omega_1 HH) + q\mathbb{X}(\omega_1 HT)\big) + q\big(p\mathbb{X}(\omega_1 TH) + q\mathbb{X}(\omega_1 TT)\big)\ . \quad (12)\end{aligned}$$

Ans. Write $\mathbb{Y} = \mathbb{E}[\mathbb{X} \mid \mathcal{F}_2]$, so that $\mathbb{Y}(\omega_1\omega_2\omega_3) = p\mathbb{X}(\omega_1\omega_2 H) + q\mathbb{X}(\omega_1\omega_2 T)$ depends only on $\omega_1\omega_2$. The atom of $\mathcal{F}_1$ containing $\omega_1\omega_2\omega_3$ is $A_{\omega_1} = \{\omega_1 HH, \omega_1 HT, \omega_1 TH, \omega_1 TT\}$ (the last two tosses are free, and we write the four outcomes by their last two tosses). Recall that, by the additive law of probabilities, we have $\mathbb{P}\{A_{\omega_1}\} = \mathbb{P}\{\omega_1\}$, so that the conditional probabilities that appear in the conditional expectation are

$$\begin{aligned}\frac{\mathbb{P}\{\omega_1 HH\}}{\mathbb{P}\{A_{\omega_1}\}} = p^2\ , \qquad & \frac{\mathbb{P}\{\omega_1 HT\}}{\mathbb{P}\{A_{\omega_1}\}} = pq\ , \\ \frac{\mathbb{P}\{\omega_1 TH\}}{\mathbb{P}\{A_{\omega_1}\}} = qp\ , \qquad & \frac{\mathbb{P}\{\omega_1 TT\}}{\mathbb{P}\{A_{\omega_1}\}} = q^2\ . \quad (13)\end{aligned}$$

The values of $\mathbb{Y}$ on these four outcomes are

$$\begin{aligned}\mathbb{Y}(\omega_1 HH) = \mathbb{Y}(\omega_1 HT) = p\mathbb{X}(\omega_1 HH) + q\mathbb{X}(\omega_1 HT)\ , \\ \mathbb{Y}(\omega_1 TH) = \mathbb{Y}(\omega_1 TT) = p\mathbb{X}(\omega_1 TH) + q\mathbb{X}(\omega_1 TT)\ .\end{aligned}$$

Hence

$$\begin{aligned}\mathbb{E}[\mathbb{Y} \mid \mathcal{F}_1](\omega_1\omega_2\omega_3) &= (p^2 + pq)\big(p\mathbb{X}(\omega_1 HH) + q\mathbb{X}(\omega_1 HT)\big) \\ &\quad + (qp + q^2)\big(p\mathbb{X}(\omega_1 TH) + q\mathbb{X}(\omega_1 TT)\big) \\ &= p\big(p\mathbb{X}(\omega_1 HH) + q\mathbb{X}(\omega_1 HT)\big) + q\big(p\mathbb{X}(\omega_1 TH) + q\mathbb{X}(\omega_1 TT)\big)\ , \quad (14)\end{aligned}$$

where, in going to the last step, we have used $p^2 + pq = p(p+q) = p$ and $qp + q^2 = q$. This is the required answer.

We can further rearrange the terms to understand the iterated conditioning better. Expanding the last formula above, we get

$$\begin{aligned}\mathbb{E}\big[\mathbb{E}[\mathbb{X} \mid \mathcal{F}_2] \mid \mathcal{F}_1\big](\omega_1\omega_2\omega_3) \\ = p^2\,\mathbb{X}(\omega_1 HH) + pq\,\mathbb{X}(\omega_1 HT) + qp\,\mathbb{X}(\omega_1 TH) + q^2\,\mathbb{X}(\omega_1 TT)\ . \quad (15)\end{aligned}$$

This expression will be useful for the next exercise. ■

Exercise 1e (2 points): Show that $\mathbb{E}[\mathbb{X} \mid \mathcal{F}_1]$ on a generic element $\omega_1\omega_2\omega_3$ of sample space evaluates to the same expression as in Exercise 1d.

Ans. The atom of $\mathcal{F}_1$ containing $\omega_1\omega_2\omega_3$ is A_{ω_1}, so $\mathbb{E}[\mathbb{X} \mid \mathcal{F}_1]$ is the direct average of $\mathbb{X}$ over its four outcomes with the weights computed above:

$$\begin{aligned}\mathbb{E}[\mathbb{X} \mid \mathcal{F}_1](\omega_1\omega_2\omega_3) &= \sum_{\omega_2'\omega_3'} \mathbb{X}(\omega_1\omega_2'\omega_3')\frac{\mathbb{P}\{\omega_1\omega_2'\omega_3'\}}{\mathbb{P}\{A_{\omega_1}\}} = \sum_{\omega_2'\omega_3'} \mathbb{X}(\omega_1\omega_2'\omega_3')\mathbb{P}\{\omega_2'\omega_3'\} \\ &= p^2\,\mathbb{X}(\omega_1 HH) + pq\,\mathbb{X}(\omega_1 HT) + qp\,\mathbb{X}(\omega_1 TH) + q^2\,\mathbb{X}(\omega_1 TT)\ . \qquad (16)\end{aligned}$$

This is identical to (15). ■

One can explain what happened above in words. When we took the conditional expectation $\mathbb{E}[\mathbb{X} \mid \mathcal{F}_2]$, its value on $\omega_1\omega_2\omega_3$ is obtained by averaging over all elements in the atom $A^{\omega_1\omega_2\omega_3}$ in $\mathcal{F}_2$ which contains it. For every fixed $\omega_1\omega_2$, there were two elements $\omega_1\omega_2 H$ and $\omega_1\omega_2 T$. Now, when we further take a conditional expectation with respect to the sub σ-algebra $\mathcal{F}_1$, its value on $\omega_1\omega_2\omega_3$ involves an averaging over all elements in the unique atom which contains $\omega_1\omega_2\omega_3$, in this case $\{\omega_1 HH, \omega_1 HT, \omega_1 TH, \omega_1 TT\}$. But note that this is in fact the union of two atoms $\{\omega_1 HH, \omega_1 HT\}$ and $\{\omega_1 TH, \omega_1 TT\}$ of $\mathcal{F}_2$. The averaging over these $\mathcal{F}_2$ atoms was already done when we computed $\mathbb{E}[\mathbb{X} \mid \mathcal{F}_2]$, and it is only the extra averaging over the two atoms that is left when we take the further conditional expectation $\mathbb{E}\big[\mathbb{E}[\mathbb{X} \mid \mathcal{F}_2] \mid \mathcal{F}_1\big]$. It is not difficult to imagine that this step-by-step averaging can be replaced by a direct averaging over all the outcomes in the atom $\{\omega_1 HH, \omega_1 HT, \omega_1 TH, \omega_1 TT\}$ of $\mathcal{F}_1$. This is basically the value of $\mathbb{E}[\mathbb{X} \mid \mathcal{F}_1]$ on $\omega_1\omega_2\omega_3$.

The same argument can be made a bit more abstractly for general σ-algebras $\mathcal{G}$ and $\mathcal{F}$ such that $\mathcal{G}$ is a sub σ-algebra of $\mathcal{F}$. The key fact to be used is that the atoms of $\mathcal{G}$ are events in $\mathcal{F}$ that are composed of certain atoms of $\mathcal{F}$. An averaging over the outcomes in an atom of $\mathcal{F}$ and then a successive average over an atom of $\mathcal{G}$ can be replaced by a direct averaging over the outcomes in the atom of $\mathcal{G}$. Note that this proof only applies to σ-algebras which are generated by their atoms. These are the type of σ-algebras that we will be interested in this course.

Exercise 1f (2 points): There is another (easier) identity involving two σ-algebras $\mathcal{G}$, $\mathcal{F}$ such that $\mathcal{G} \subseteq \mathcal{F}$. This is

$$\mathbb{E}\big[\mathbb{E}[\mathbb{X} \mid \mathcal{G}] \mid \mathcal{F}\big] = \mathbb{E}[\mathbb{X} \mid \mathcal{G}]\ . \qquad (17)$$

Show this for the choice $\mathcal{G} = \mathcal{F}_1$ and $\mathcal{F} = \mathcal{F}_2$ by showing that the value of both sides are the same on a generic element $\omega_1\omega_2\omega_3$ of the sample space Ω_3. ■

Ans. Let $\mathbb{Z} = \mathbb{E}[\mathbb{X} \mid \mathcal{F}_1]$. Its value on $\omega_1\omega_2\omega_3$ is given by (Exercise 1e) the explicit expression, which depends only on ω_1. So $\mathbb{Z}(\omega_1\omega_2\omega_3)$ is constant on each atom A_{ω_1} and is dependent only on ω_1. We denote this property as

$$\mathbb{Z}(\omega_1 HH) = \mathbb{Z}(\omega_1 HT) = \mathbb{Z}(\omega_1 TH) = \mathbb{Z}(\omega_1 TT) = \mathbb{Z}(\omega_1 \cdots)\ , \qquad (18)$$

where the $\cdots$ is to indicate that $\mathbb{Z}$ does not depend on the other coin tosses.

Then, we have $\mathbb{E}\big[\mathbb{E}[\mathbb{X} \mid \mathcal{F}_1] \mid \mathcal{F}_2\big] = \mathbb{E}[\mathbb{Z} \mid \mathcal{F}_2]$ which, on a generic element $\omega_1\omega_2\omega_3$, is

$$\begin{aligned}\mathbb{E}\big[\mathbb{E}[\mathbb{X} \mid \mathcal{F}_1] \mid \mathcal{F}_2\big](\omega_1\omega_2\omega_3) &= p\,\mathbb{Z}(\omega_1\omega_2 H) + q\,\mathbb{Z}(\omega_1\omega_2 T) = (p+q)\,\mathbb{Z}(\omega_1\cdots) \\ &= \mathbb{Z}(\omega_1\cdots) = \mathbb{E}[\mathbb{X} \mid \mathcal{F}_1](\omega_1\omega_2\omega_3)\ . \end{aligned} \tag{19}$$

The two sides agree on every element of Ω_3. ■

2 The risk-neutral pricing formula for derivative securities

The stock price follows the binomial tree model

$$\mathbb{S}_n(\omega_1\cdots\omega_n) = \begin{cases} u\,\mathbb{S}_{n-1}(\omega_1\cdots\omega_{n-1}) & \text{when } \omega_n = H\ , \\ d\,\mathbb{S}_{n-1}(\omega_1\cdots\omega_{n-1}) & \text{when } \omega_n = T\ . \end{cases} \tag{20}$$

The money market account offers a risk-free rate of interest r per period of the binomial tree model. The no-arbitrage pricing formula for the price process $\mathbb{V}_n$ of a European-style derivative security with expiration time N is given by

$$\frac{\mathbb{V}_n}{(1+r)^n} = \widetilde{\mathbb{E}}\left[\frac{\mathbb{V}_N}{(1+r)^N}\ \middle|\ \mathcal{F}_n\right]\ , \tag{21}$$

where $\mathbb{V}_N$ is the payoff the derivative security at expiration time N which depends on the stock price process in some way, $\mathcal{F}_n$ is the σ-algebra determined by the first n coin tosses, and $\widetilde{\mathbb{E}}$ is the conditional expectation with respect to the risk-neutral probabilities. Note that the price $\mathbb{V}_n$ at time n depends only on the first n coin tosses due to the conditional expectation with respect to the σ-algebra $\mathcal{F}_n$.

Exercise 2a (2 points): Recall that the expressions for the risk-neutral probabilities $\tilde{p}$ and $\tilde{q}$ are derived by imposing that the discounted stock price process is a martingale with respect to the risk-neutral probabilities. This is the condition

$$\frac{\mathbb{S}_n(\omega_1\cdots\omega_n)}{(1+r)^n} = \widetilde{\mathbb{E}}\left[\frac{\mathbb{S}_{n+1}}{(1+r)^{n+1}}\ \middle|\ \mathcal{F}_n\right](\omega_1\cdots\omega_n)\ . \tag{22}$$

Compute the conditional expectation on the right hand side by averaging over the remaining $(n+1)^{\text{th}}$ coin toss and derive the expressions for $\tilde{p}$ and $\tilde{q}$. Note that since u, d and r remain the same for every time period, the risk-neutral probabilities are also the same for any n in the above formula. ■

Ans. Given $\omega_1\cdots\omega_n$, the only uncertainty in $\mathbb{S}_{n+1}$ is the $(n+1)^{\text{th}}$ toss, with

$$\begin{aligned}\mathbb{S}_{n+1}(\omega_1\cdots\omega_n H) &= u\,\mathbb{S}_n(\omega_1\cdots\omega_n)\ , \\ \mathbb{S}_{n+1}(\omega_1\cdots\omega_n T) &= d\,\mathbb{S}_n(\omega_1\cdots\omega_n)\ . \end{aligned}$$

Averaging with the risk-neutral probabilities (as in Exercise 1c, with $p \to \tilde{p}$, $q \to \tilde{q}$) and pulling out the deterministic factor $(1+r)^{-(n+1)}$,

$$\widetilde{\mathbb{E}}\left[\frac{\mathbb{S}_{n+1}}{(1+r)^{n+1}} \;\middle|\; \mathcal{F}_n\right](\omega_1\cdots\omega_n) = \frac{\tilde{p}\,u\,\mathbb{S}_n(\omega_1\cdots\omega_n) + \tilde{q}\,d\,\mathbb{S}_n(\omega_1\cdots\omega_n)}{(1+r)^{n+1}} \;. \tag{23}$$

The martingale condition then reads

$$\frac{\mathbb{S}_n}{(1+r)^n} = \frac{\mathbb{S}_n\,(\tilde{p}\,u + \tilde{q}\,d)}{(1+r)^{n+1}} \quad \Longrightarrow \quad \tilde{p}\,u + \tilde{q}\,d = 1 + r \;, \tag{24}$$

where $\mathbb{S}_n > 0$ was cancelled and the dependence on n and on $\omega_1\cdots\omega_n$ drops out. Together with $\tilde{p} + \tilde{q} = 1$, i.e., $d + \tilde{p}\,(u-d) = 1 + r$, we get

$$\tilde{p} = \frac{1+r-d}{u-d} \;, \qquad \tilde{q} = \frac{u-1-r}{u-d} \;. \tag{25}$$

Both are positive when $d < 1 + r < u$ giving legitimate probabilities. Recall that the above relation between u, d and r is what ensures no arbitrage in trades involving the stock and the money market. ■

Exercise 2b (2 points): Using the iterated conditioning property with $\mathcal{G} = \mathcal{F}_n$ and $\mathcal{F} = \mathcal{F}_{n+1}$, derive the formula

$$\mathbb{V}_n(\omega_1\cdots\omega_n) = \frac{1}{1+r}\widetilde{\mathbb{E}}[\mathbb{V}_{n+1} \mid \mathcal{F}_n](\omega_1\cdots\omega_n) \;. \tag{26}$$

Ans. Start from the pricing formula at time n and insert the pricing formula at time $n+1$, $\mathbb{V}_{n+1}/(1+r)^{n+1} = \widetilde{\mathbb{E}}\big[\mathbb{V}_N/(1+r)^N \mid \mathcal{F}_{n+1}\big]$. Since $\mathcal{F}_n \subseteq \mathcal{F}_{n+1}$,

$$\begin{aligned}\frac{\mathbb{V}_n}{(1+r)^n} &= \widetilde{\mathbb{E}}\left[\frac{\mathbb{V}_N}{(1+r)^N} \;\middle|\; \mathcal{F}_n\right] = \widetilde{\mathbb{E}}\left[\widetilde{\mathbb{E}}\left[\frac{\mathbb{V}_N}{(1+r)^N} \;\middle|\; \mathcal{F}_{n+1}\right] \;\middle|\; \mathcal{F}_n\right] = \widetilde{\mathbb{E}}\left[\frac{\mathbb{V}_{n+1}}{(1+r)^{n+1}} \;\middle|\; \mathcal{F}_n\right] \\ &= \frac{1}{(1+r)^{n+1}}\,\widetilde{\mathbb{E}}[\mathbb{V}_{n+1} \mid \mathcal{F}_n] \;,\end{aligned} \tag{27}$$

where the second equality is the iterated conditioning property with $\mathcal{G} = \mathcal{F}_n$, $\mathcal{F} = \mathcal{F}_{n+1}$, and in the last step we pulled out the non-random factor $(1+r)^{-(n+1)}$. Multiplying by $(1+r)^n$ gives $\mathbb{V}_n = \frac{1}{1+r}\widetilde{\mathbb{E}}[\mathbb{V}_{n+1} \mid \mathcal{F}_n]$. ■

Exercise 2c (2 points): Recall that the derivative price $\mathbb{V}_n$ on a generic outcome in sample space $\omega_1\cdots\omega_N$ is dependent only on the first n coin toss outcomes $\omega_1\cdots\omega_n$. Show that this is given by

$$\mathbb{V}_n(\omega_1\cdots\omega_n) = \frac{1}{1+r}\big[\tilde{p}\mathbb{V}_{n+1}(\omega_1\cdots\omega_n H) + \tilde{q}\mathbb{V}_{n+1}(\omega_1\cdots\omega_n T)\big] \;. \tag{28}$$

In class, we interpreted the above formula in terms of a pictorial rule for calculating derivative prices on a binomial tree. Just like we have a binomial tree for stocks, we can draw a binomial

tree for the derivative price $\mathbb{V}_n$. A sequence of n coin tosses $\omega_1 \cdots \omega_n$ leads us to a particular node at time n of the binomial tree that is reached by following the outcomes ω_1, ω_2,..., ω_n along the tree from time 0 to time n. This node is associated with the price $\mathbb{V}_n(\omega_1 \cdots \omega_n)$. The above formula for $\mathbb{V}_n$ in terms of $\mathbb{V}_{n+1}$ indicates that $\mathbb{V}_n$ at a particular node can be computed by taking the average (with respect to the risk-neutral probabilities) of the two $\mathbb{V}_{n+1}$ prices that emanate out of that node, and then discounting the result with $1/(1+r)$.

Ans. By Exercise 2b, $\mathbb{V}_n = \frac{1}{1+r}\widetilde{\mathbb{E}}[\mathbb{V}_{n+1} \mid \mathcal{F}_n]$. The computation of Exercise 1c (the conditional expectation given the first n tosses averages over the $(n+1)^{\text{th}}$ toss, whose atom is $\{\omega_1 \cdots \omega_n H, \omega_1 \cdots \omega_n T\}$) goes through with the risk-neutral probabilities $\tilde{p}, \tilde{q}$ in place of p, q and $\mathbb{X} = \mathbb{V}_{n+1}$:

$$\widetilde{\mathbb{E}}[\mathbb{V}_{n+1} \mid \mathcal{F}_n](\omega_1 \cdots \omega_n \cdots) = \tilde{p}\, \mathbb{V}_{n+1}(\omega_1 \cdots \omega_n H) + \tilde{q}\, \mathbb{V}_{n+1}(\omega_1 \cdots \omega_n T) \ . \tag{29}$$

The right hand side depends only on $\omega_1 \cdots \omega_n$, and dividing by $1+r$ gives the required formula. ■

Exercise 2d (3 points): Consider a European put option with expiration time $N = 3$ and strike price $K = 4$. The payoff of the option is $\mathbb{V}_3 = (K - \mathbb{S}_3)^+$. Compute the risk-neutral price of the put option at all prior time periods from $n = 0$ to $n = 2$ for all possible *stock price scenarios*, i.e., sequences of coin tosses. This is called the risk-neutral price evolution of the put option. Show the risk-neutral prices on a 3-period binomial tree. Take $\mathbb{S}_0 = 4$, $u = 2$, $d = u^{-1} = 1/2$, and $r = 1/4$. ■

Ans. The risk-neutral probabilities (Exercise 2a) are

$$\tilde{p} = \frac{1 + \frac{1}{4} - \frac{1}{2}}{2 - \frac{1}{2}} = \frac{1}{2} \ , \qquad \tilde{q} = \frac{1}{2} \ , \tag{30}$$

so the backward recursion of Exercise 2c is

$$\begin{aligned} \mathbb{V}_n(\omega_1 \cdots \omega_n) &= \frac{1}{1.25}\left(\frac{1}{2}\mathbb{V}_{n+1}(\omega_1 \cdots \omega_n H) + \frac{1}{2}\mathbb{V}_{n+1}(\omega_1 \cdots \omega_n T)\right) \ , \\ &= 0.4\left(\mathbb{V}_{n+1}(\omega_1 \cdots \omega_n H) + \mathbb{V}_{n+1}(\omega_1 \cdots \omega_n T)\right) \ . \end{aligned} \tag{31}$$

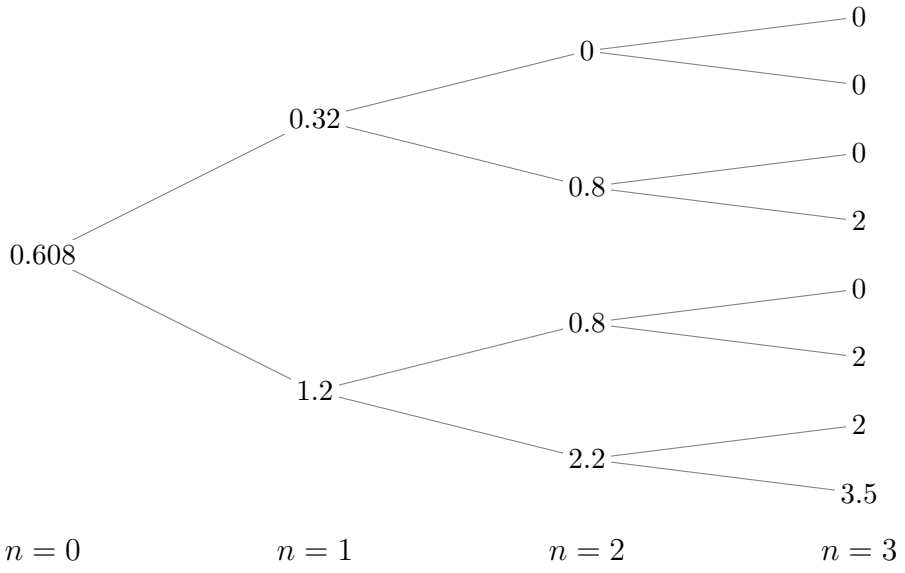
The terminal values are

$\omega_1\omega_2\omega_3$	HHH	HHT	HTH	HTT	THH	THT	TTH	TTT
$\mathbb{S}_3$	32	8	8	2	8	2	2	0.5
$\mathbb{V}_3 = (4 - \mathbb{S}_3)^+$	0	0	0	2	0	2	2	3.5

Working backwards,

$$\begin{aligned}
\mathbb{V}_2(HH) &= 0.4(0+0) = 0\ , & \mathbb{V}_2(HT) &= 0.4(0+2) = 0.8\ ,\\
\mathbb{V}_2(TH) &= 0.4(0+2) = 0.8\ , & \mathbb{V}_2(TT) &= 0.4(2+3.5) = 2.2\ ,\\
\mathbb{V}_1(H) &= 0.4(0+0.8) = 0.32\ , & \mathbb{V}_1(T) &= 0.4(0.8+2.2) = 1.2\ ,\\
\mathbb{V}_0 &= 0.4(0.32+1.2) = 0.608\ . & & \qquad (32)
\end{aligned}$$

As a check, we can compute $\mathbb{V}_0$ directly using the formula $\mathbb{V}_0 = \widetilde{\mathbb{E}}[\mathbb{V}_N \mid \mathcal{F}_0]/(1+r)^N = \widetilde{\mathbb{E}}[\mathbb{V}_N]/(1+r)^N$. This gives $\mathbb{V}_0 = \widetilde{\mathbb{E}}[\mathbb{V}_3]/(1+r)^3 = \frac{1}{1.25^3}(\frac{1}{8}2+\frac{1}{8}2+\frac{1}{8}2+\frac{1}{8}3.5) = 1.1875/1.953125 = 0.608$. The risk-neutral price evolution is:



Note that the tree actually recombines since the prices at two different nodes are equal, e.g., $\mathbb{V}_2(HT) = \mathbb{V}_2(TH)$, because $\mathbb{V}_3$ depends only on $\mathbb{S}_3$. Thus, we could have drawn a single node with the price 0.8 at time 2 instead of the two nodes shown above. ■

Exercise 2e (3 points): Analyse the put option prices $\mathbb{V}_n$ when the strike price K is much higher than the highest stock price at time 3 ("in the money", ITM) and when K is much lower than the lowest possible stock price at time 3 ("out of the money", OTM). What trends do you observe? You can choose explicit values for K for the ITM and OTM cases if you want and then answer the question. ■

Ans. We continue with the same market parameters as in Exercise 2d. That is, $\mathbb{S}_0 = 2$, $u = d^{-1} = 2$, $r = 0.25$ which results in the final stock price lying in the range $\mathbb{S}_3 \in [0.5, 32]$ and the risk-neutral probabilities $\tilde{p} = \tilde{q} = \frac{1}{2}$.

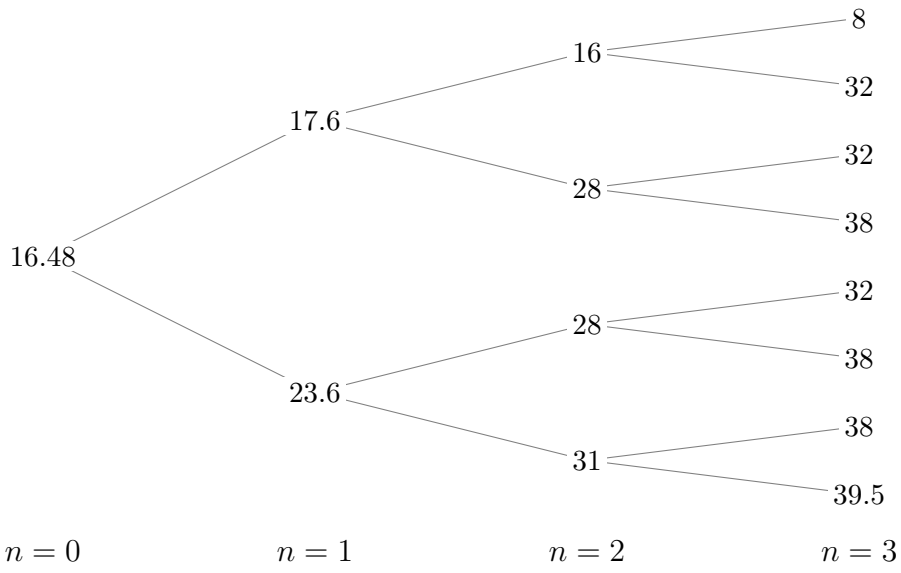
The ITM case: When K is greater than 32, the payoff function $(K - \mathbb{S}_3)^+$ is always equal to $K - \mathbb{S}_3$, .i.e., the + part is always active. Then it is easy to use the martingale property

of $\mathbb{S}_n$ under risk-neutral probabilities to get

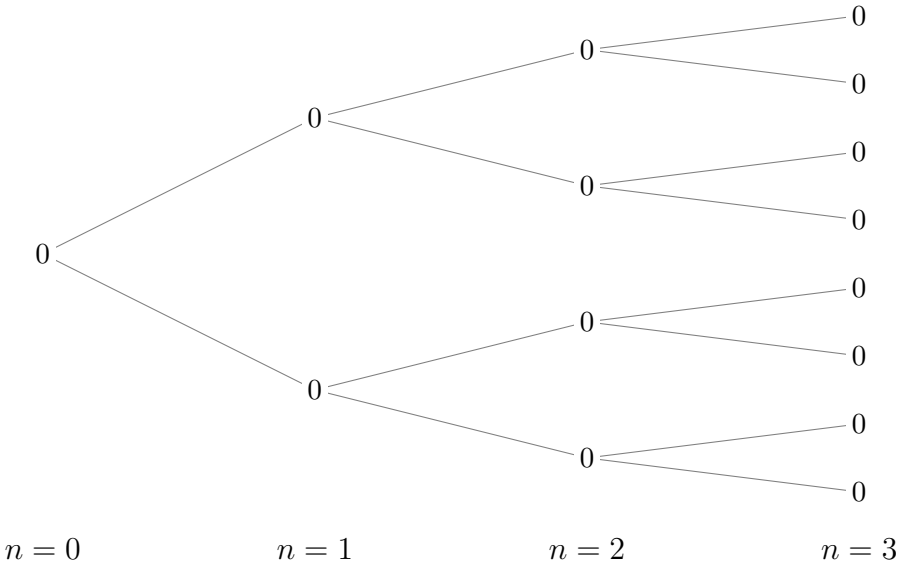
$$\frac{\mathbb{V}_n}{(1+r)^n} = \widetilde{\mathbb{E}}\left[\frac{K-\mathbb{S}_3}{(1+r)^3}\,\middle|\,\mathcal{F}_n\right] = \frac{K}{(1+r)^3} - \frac{\mathbb{S}_n}{(1+r)^n} \quad \Longrightarrow \quad \mathbb{V}_n = \frac{K}{(1+r)^{3-n}} - \mathbb{S}_n\ . \quad (33)$$

Thus, in this case, the price $\mathbb{V}_n$ is linear in the strike price K as well as the stock price $\mathbb{S}_n$. The option price at time n increases if $\mathbb{S}_n$ decreases, while it increases as K increases.

With $K = 40$ the put option price process is



OTM case: If K is lesser than the smallest stock price at time 3, i.e., $K < 0.5$, then the payoff of the put option is 0 for every scenario of the stock price at time 3. This means that the rolled back put option prices are also zero at all times prior to expiry. Hence, if the strike price is low enough, the put option is worthless.



Thus, generally, a European put option is expensive if the strike price is very high and is

worthless if the strike is very low. ■

Note that the European call and put options that we have considered so far have payoffs which depended only on the stock price at the time of expiration. Options whose payoffs depend not only on the final stock price but stock prices at some prior times as well are called **exotic options**. The risk-neutral pricing formula also applies to such options as long as the option can be exercised only at the time of expiration and not earlier, i.e., they are European-style. We look at the risk-neutral pricing of two such options next.

Exercise 2f (3 points): Take the same financial market parameters u, d and r as in Exercise 2c. Compute the risk-neutral price evolution of the *Asian call option* with expiration time $N = 3$ and strike $K = 5$. Set $\mathbb{Y}_n = \frac{1}{n+1}\sum_{k=0}^{n}\mathbb{S}_k$ to be the stock price average random process. The payoff of the option is given by

$$\mathbb{V}_N = (\mathbb{Y}_N - K)^+ \ . \tag{34}$$

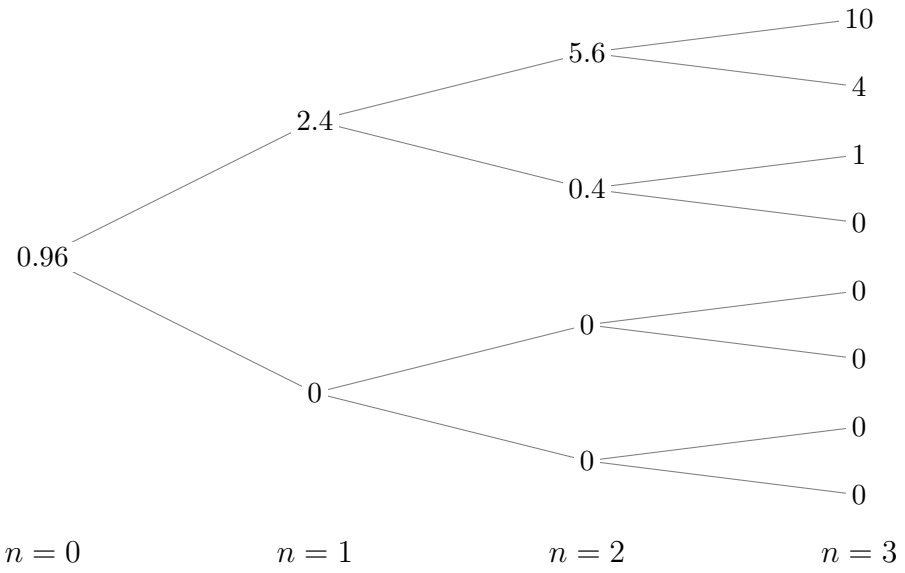
Show the risk-neutral prices on a 3-period binomial tree. ■

Ans. The payoff depends on the whole path, so we tabulate $\mathbb{Y}_3$ path by path:

$\omega_1\omega_2\omega_3$	$(\mathbb{S}_0, \mathbb{S}_1, \mathbb{S}_2, \mathbb{S}_3)$	$\sum_{k=0}^{3}\mathbb{S}_k$	$\mathbb{Y}_3$	$\mathbb{V}_3 = (\mathbb{Y}_3 - 5)^+$
HHH	$(4, 8, 16, 32)$	60	15	10
HHT	$(4, 8, 16, 8)$	36	9	4
HTH	$(4, 8, 4, 8)$	24	6	1
HTT	$(4, 8, 4, 2)$	18	4.5	0
THH	$(4, 2, 4, 8)$	18	4.5	0
THT	$(4, 2, 4, 2)$	12	3	0
TTH	$(4, 2, 1, 2)$	9	2.25	0
TTT	$(4, 2, 1, 0.5)$	7.5	1.875	0

Applying the recursion formula (31) node by node, we get

$$\begin{aligned}
\mathbb{V}_2(HH) &= 0.4(10+4) = 5.6 \ , & \mathbb{V}_2(HT) &= 0.4(1+0) = 0.4 \ , \\
\mathbb{V}_2(TH) &= 0.4(0+0) = 0 \ , & \mathbb{V}_2(TT) &= 0.4(0+0) = 0 \ , \\
\mathbb{V}_1(H) &= 0.4(5.6+0.4) = 2.4 \ , & \mathbb{V}_1(T) &= 0.4(0+0) = 0 \ , \\
\mathbb{V}_0 &= 0.4(2.4+0) = 0.96 \ .
\end{aligned} \tag{35}$$



Unlike the put of Exercise 2d, the tree does *not* recombine: $\mathbb{V}_2(HT) = 0.4 \neq \mathbb{V}_2(TH) = 0$ even though $\mathbb{S}_2(HT) = \mathbb{S}_2(TH) = 4$, because the running average $\mathbb{Y}_2$ differs (16/3 vs. 10/3). The price $\mathbb{V}_n$ depends on the full path up to time n, not only on the stock price at time n. ■

Exercise 2g (3 points): Consider the same financial market comprising of one stock with $\mathbb{S}_0 = 4$, $u = d^{-1} = 2$ and a money market account with risk-free rate of interest $r = 1/4$. Compute the risk-neutral price evolution of the *lookback option* with expiration time $N = 3$. Let $\mathbb{M}_n = \max_{0 \leq k \leq n} \mathbb{S}_n$ be the maximum-to-date process for the stock price process, i.e., $\mathbb{M}_n$ is the maximum price of the stock up to the current time n. The payoff of the lookback option at the time of expiration is

$$\mathbb{V}_N = \mathbb{M}_N - \mathbb{S}_N \ . \tag{36}$$

Show the risk-neutral prices on a 3-period binomial tree. ■

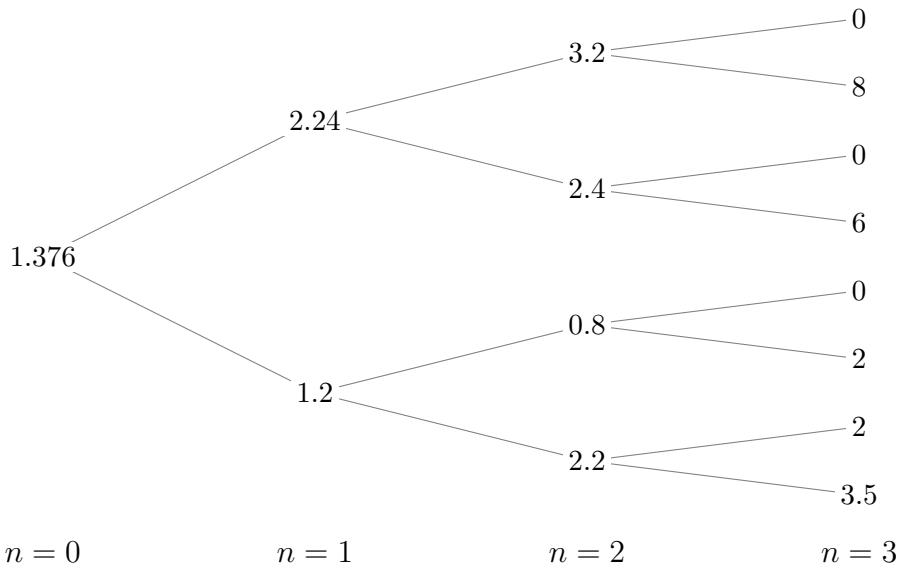
Ans. First, we compute the payoffs for the different stock price scenarios:

$\omega_1\omega_2\omega_3$	$(\mathbb{S}_0, \mathbb{S}_1, \mathbb{S}_2, \mathbb{S}_3)$	$\mathbb{M}_3$	$\mathbb{S}_3$	$\mathbb{V}_3 = \mathbb{M}_3 - \mathbb{S}_3$
HHH	$(4, 8, 16, 32)$	32	32	0
HHT	$(4, 8, 16, 8)$	16	8	8
HTH	$(4, 8, 4, 8)$	8	8	0
HTT	$(4, 8, 4, 2)$	8	2	6
THH	$(4, 2, 4, 8)$	8	8	0
THT	$(4, 2, 4, 2)$	4	2	2
TTH	$(4, 2, 1, 2)$	4	2	2
TTT	$(4, 2, 1, 0.5)$	4	0.5	3.5

Again, the risk-neutral probabilities are $\tilde{p} = \tilde{q} = 1/2$ and we may ue the recursion formula

(31). Working backwards, we have

$$\begin{aligned}
\mathbb{V}_2(HH) &= 0.4(0+8) = 3.2\ , & \mathbb{V}_2(HT) &= 0.4(0+6) = 2.4\ ,\\
\mathbb{V}_2(TH) &= 0.4(0+2) = 0.8\ , & \mathbb{V}_2(TT) &= 0.4(2+3.5) = 2.2\ ,\\
\mathbb{V}_1(H) &= 0.4(3.2+2.4) = 2.24\ , & \mathbb{V}_1(T) &= 0.4(0.8+2.2) = 1.2\ ,\\
\mathbb{V}_0 &= 0.4(2.24+1.2) = 1.376\ . & & \qquad (37)
\end{aligned}$$



As for the Asian option, the price is path dependent. For instance, $\mathbb{V}_2(HT) = 2.4 \neq \mathbb{V}_2(TH) = 0.8$ although $\mathbb{S}_2(HT) = \mathbb{S}_2(TH) = 4$, because the maximum-to-date differs in the two scenarios ($\mathbb{M}_2(HT) = 8$, $\mathbb{M}_2(TH) = 4$). ■